

Manual of High-Speed Barrier Gate Mechanism and Control Driver

OB-BCLV-V1.02

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1 Warnings

1.1 Safety Warnings

Important! This manual contains important safety-related information. Improper installation may result in serious safety accidents. Please read this manual thoroughly before installation. If you have any questions in installation, stop and contact the opener bear technical service center.

Important! Please keep it properly for future maintenance reference.

1.2 Installation Warnings

- Caution! Read the manual before use and operate in accordance with requirements.
- Packaging materials with potential hazards must be kept out of the reach of children.
- Keep the manual safely for future use.
- This product is designed and manufactured strictly in accordance with the operating guidelines in the document. Any use or operation not in compliance with the guidelines may cause product damage or hazards.
- The opener bear shall not be liable for any adverse consequences arising from improper use or use beyond the specified scope.
- Do not install this product in explosion-hazardous areas; flammable gases pose serious safety risks.
- Mechanical parts must comply with relevant national standards.
- Supporting products must be designed and manufactured in accordance with the manual's requirements.
- The opener bear shall not be liable for problems caused by neglecting process requirements for precision components during construction or causing deformation of such components.
- Installation must comply with relevant national standards.
- Switch off the power before product maintenance.
- Check whether earthing is correct.
- Safety devices (e.g. magnetic switches, sensors, etc.) shall be protected against damage.
- The opener bear shall not be liable for safety issues or malfunctions arising from supporting products not provided by the opener bear.
- Only original parts shall be repaired.
- Do not make any modifications to components of the system.
- The service staff must fully explain the operation methods and emergency regulations for the user.
- Keep children and other unrelated personnel away during product installation.
- Keep remote controls out of the reach of children to prevent accidents.
- Any operation not specified in the manual is prohibited.

1.3 Usage Warnings

- This product is not suitable for use by children, persons with disabilities, or persons without relevant knowledge and experience.
- Children, the elderly and pregnant women must be supervised when passing through to avoid accidental injury caused by contact.
- Prevent children from touching motors and control equipment, and keep remote controller out of their reach.

2 Product Information

2.1 Application Scope

The high-speed barrier gate mechanism and control driver is an upgraded product developed by the opener bear. It is an electromechanical integration product composed of a servo reduction motor and a controller. The controller realizes more precise control over the motor, and the product have excellent electrical performance.

The reduction sinusoidal servo motor drive controller features large starting torque, extremely low temperature rise, more precise control, stable operation, low noise and high safety.

2.2 Main Technical Parameters

Item	Specification	Remark
Rated operating voltage	24VDC	
Motor operating voltage	24VDC	
Rated power of single motor	300W	
Rated operating current of motor	12.5A	
Motor speed	3000rpm	
MCBF	≥2,000,000 cycles	
Dimensions	300(L)*270(D)*616(H)mm	
Applicable arm length	≤6m(straight arm), 5m(fence arm)	
Life-cycles	≥10,000,000 cycles	
Single-trip running time	0.9–6s,adjustable(aluminum arm) 0.7–6s,adjustable(carbon fiber arm)	
Temperature rise	≤15℃	
Operating noise	≤60dB	
Controller dimensions	176.6(H)*103(W)*25(D)mm	
Altitude	≤2000m	
Operating temperature	-35-70℃	
Operating humidity	≤95%, non-condensing	

3 Main Functions and Performance Description

3.1 Product Features

- Equipped with self-diagnosis and alarm functions for ease maintenance and use.
- Connect to control signals from any external control device, push buttons or remote controls to realize barrier up and down control.
- Automatic reset function: If no vehicle passage is detected within the time domain while the barrier is open, it will automatically return to the close state.
- Light indication: High-brightness passage lights indicate status and guide passage.
- Fire input: Connectable with fire alarm signal to keep the barrier normal open without power failure.
- Alarm function: An alarm will be triggered automatically when the barrier arm is forced open .
- Adjustable speed: Multi-level adjustment running time, configurable by users according to needs.
- New-generation industrial-grade ARM control system equip with additional functions such as multi-function digital setting, encryption and reset.
- Anti-hit function: During barrier arm closing, if a loop detector signal or anti-hit signal is detected, the arm will stop or rise to the open state according to the set “anti-hit action” parameters.
- Obstacle rebound function: If the barrier arm encounter resistance during closing and the force and duration meet the domain, the barrier will stop or rise to the open state based on the “descending obstacle action” parameters.
- Counting function: After receiving multiple open signals, the barrier arm will close automatically when corresponding vehicles pass. If the number of pass vehicles is less than the number of received open signals, the automatic reset function will be activated.
- Power-on close & power-off open: The barrier arm opens automatically after power off and return to close state automatically after power on.
- Light control function: The barrier controls light changes according to arm movement status to guide drivers.
- Anti-freezing function: When enabled, the barrier can operate in anti-freezing mode.
- Long-duration open signal alarm: An alarm will be issued if the open signal hold and exceeds the domain.

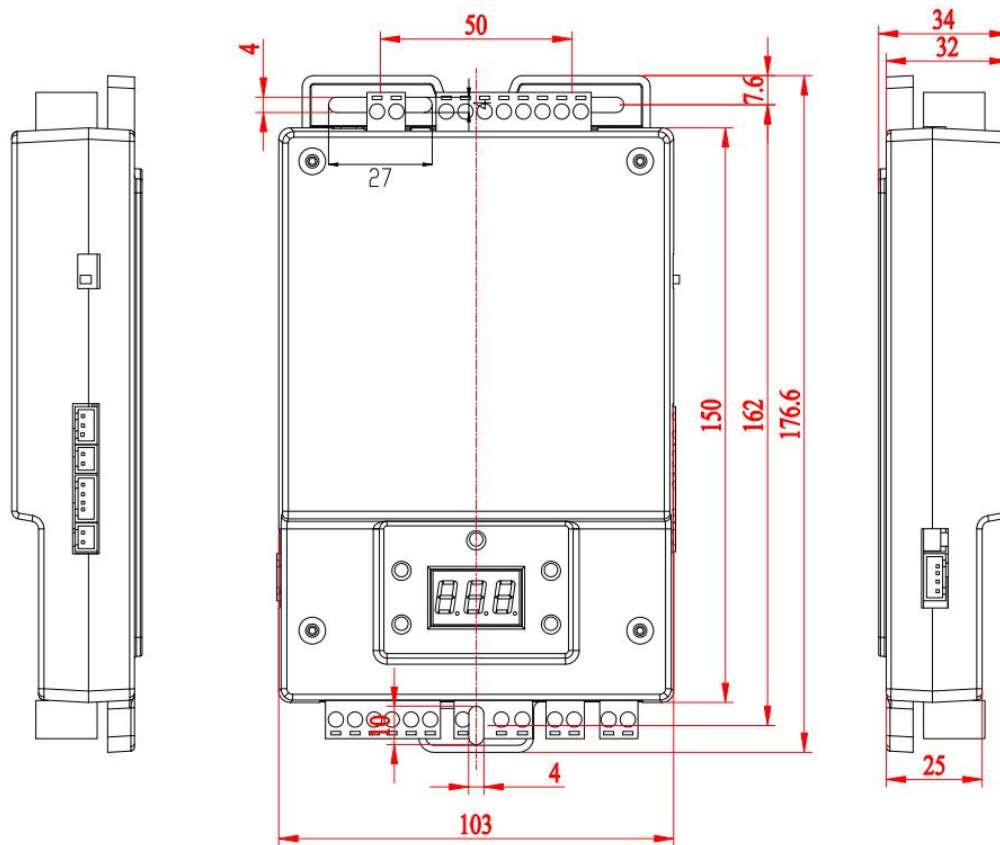
3.2 Main Product Characteristics

- 3 times overload capacity.
- WeChat Mini Program enables remote control, fault reporting, and application of servo control algorithms.
- Low motor temperature rise.

4 Installation Instructions

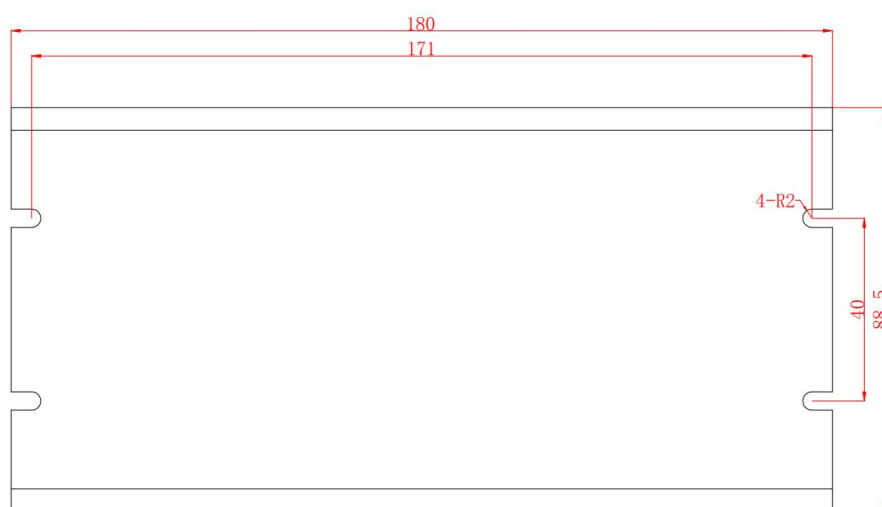
Install the mechanism control driver according to the mounting and positioning holes of product. The control driver is described as follows:

4.1 Controller Installation Drawing



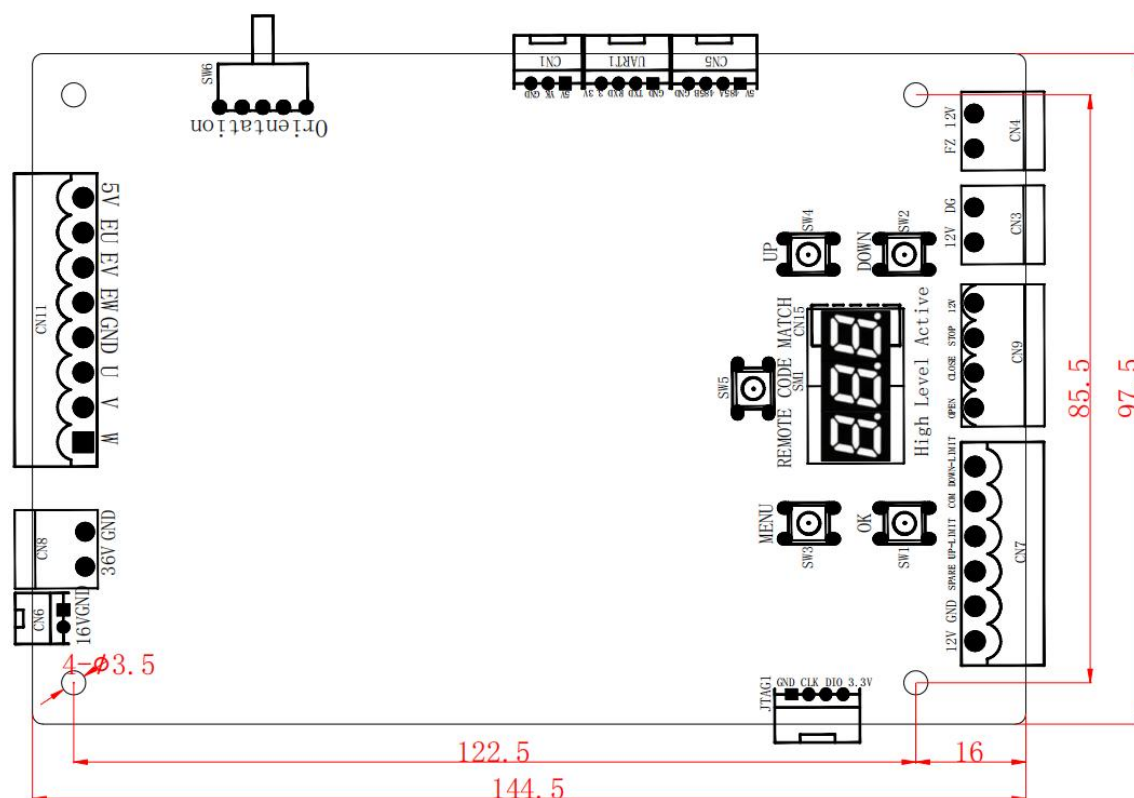
Controller Installation Dimension

4.2 Switching Power Supply Installation Dimension diagram



Switching Power Supply Installation Dimension

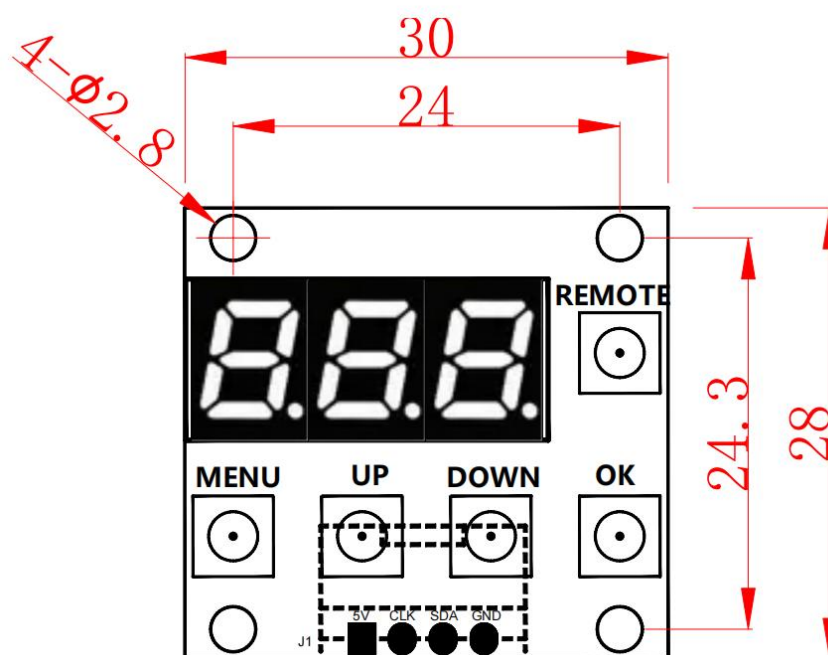
4.3 Low-Voltage Mainboard Installation Dimension Drawing



Low-Voltage Mainboard Installation dimension

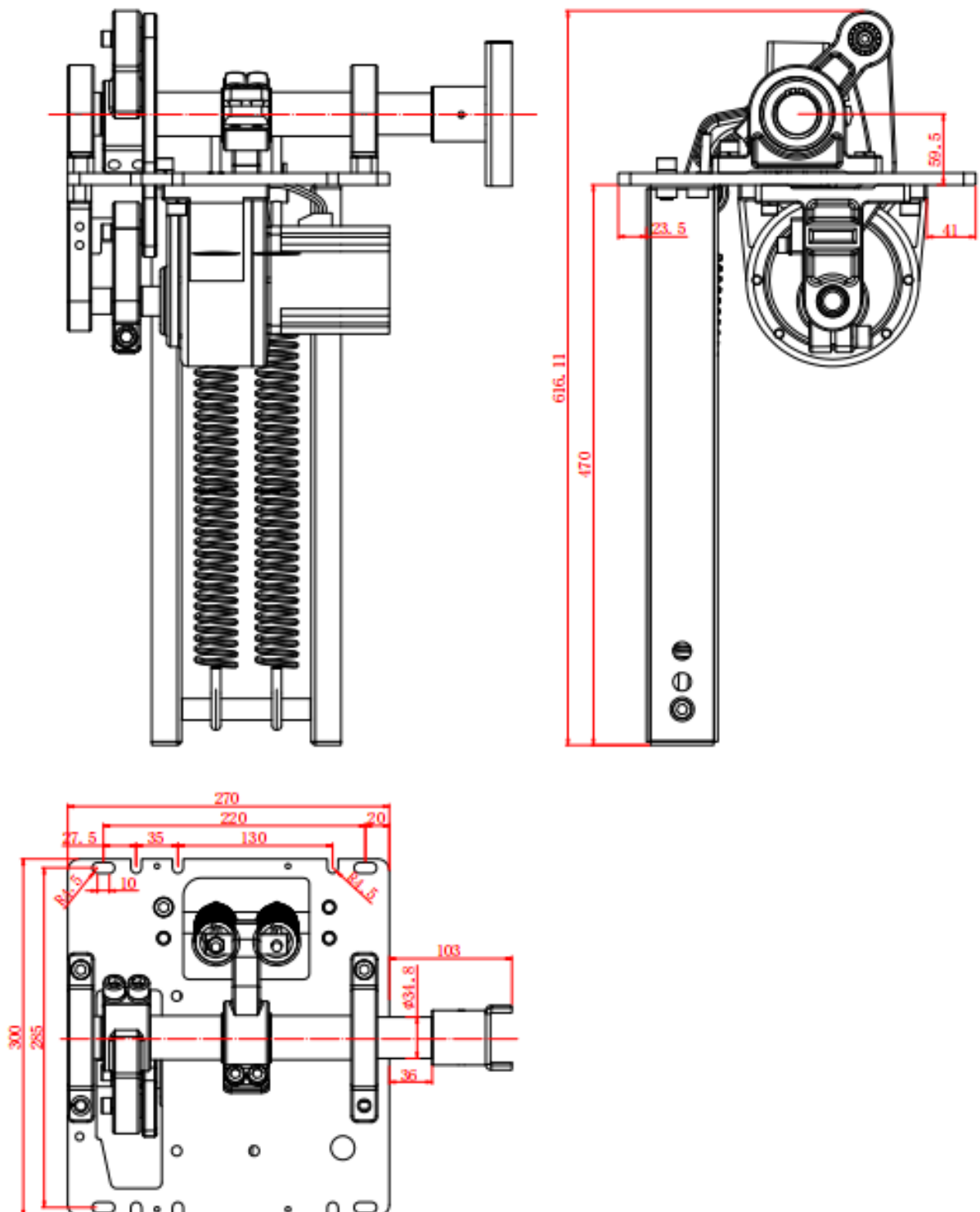
Note: 4- ϕ 3.5 refers to mounting holes.

4.4 Display Board Installation Dimension Drawing

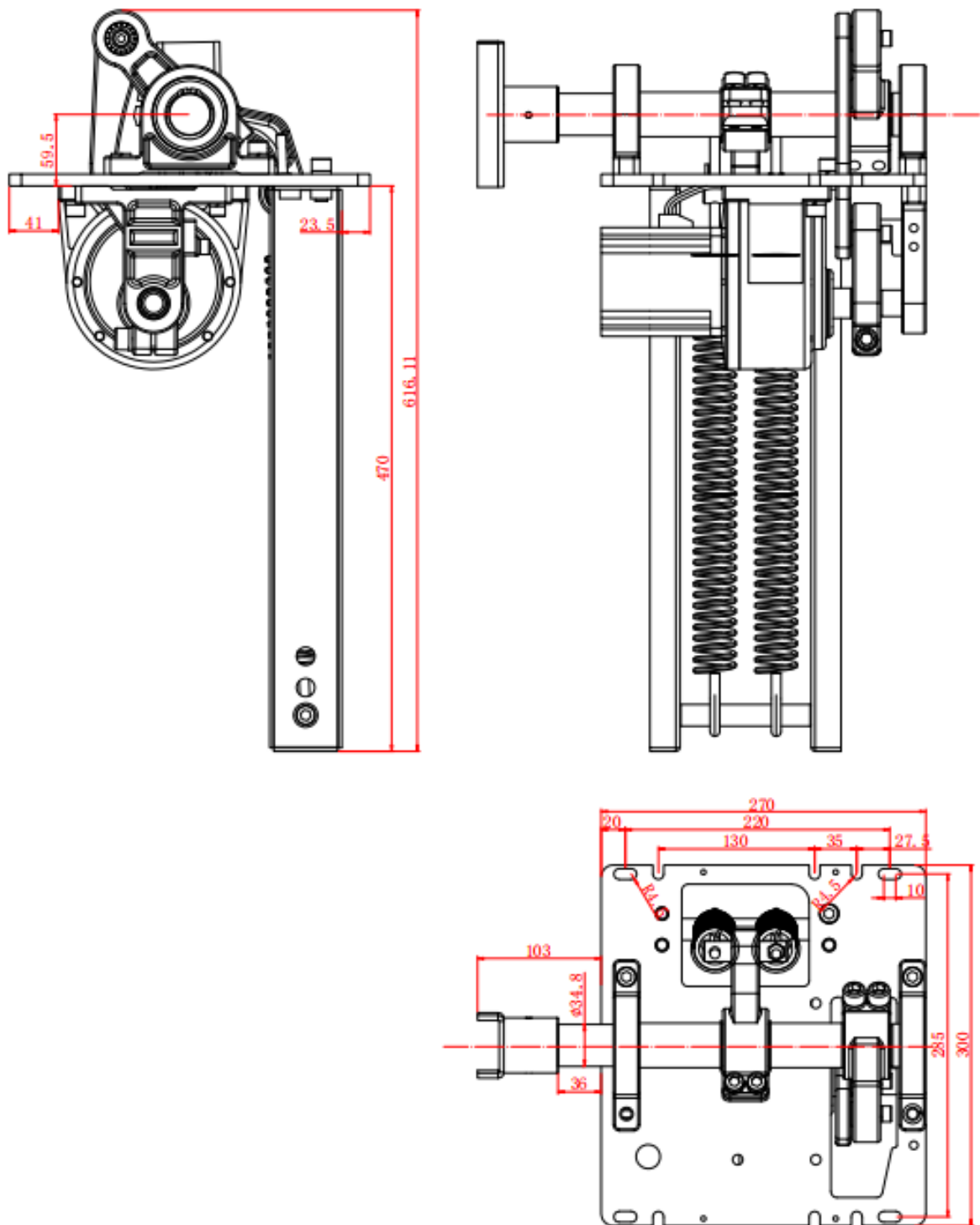


Display Board Installation Dimension

4.5 Mechanism Installation Dimension Drawing



Right-direction Mechanism Installation Dimension

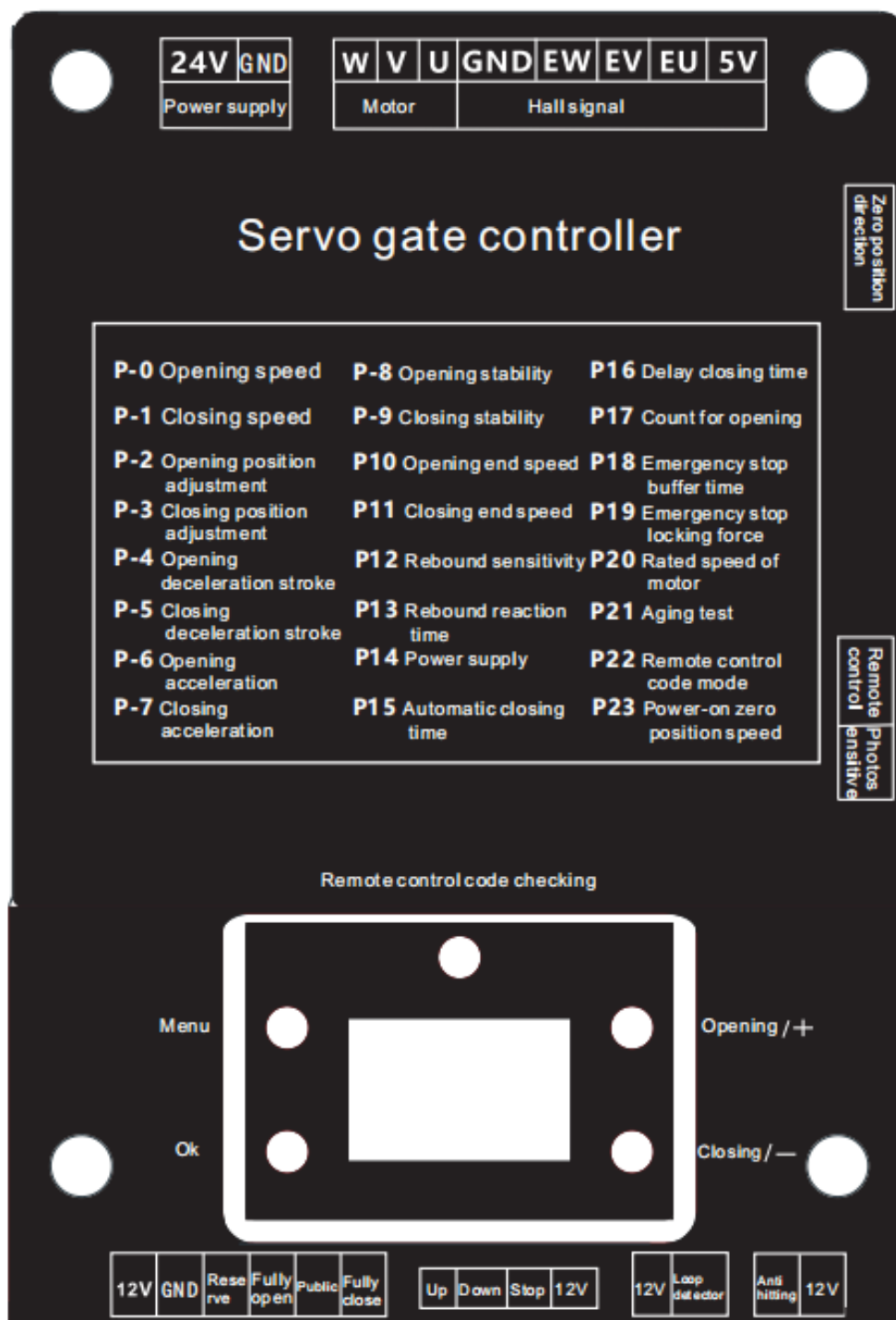


Left-direction Mechanism Installation Dimension

5 Power-On and Mainboard Description

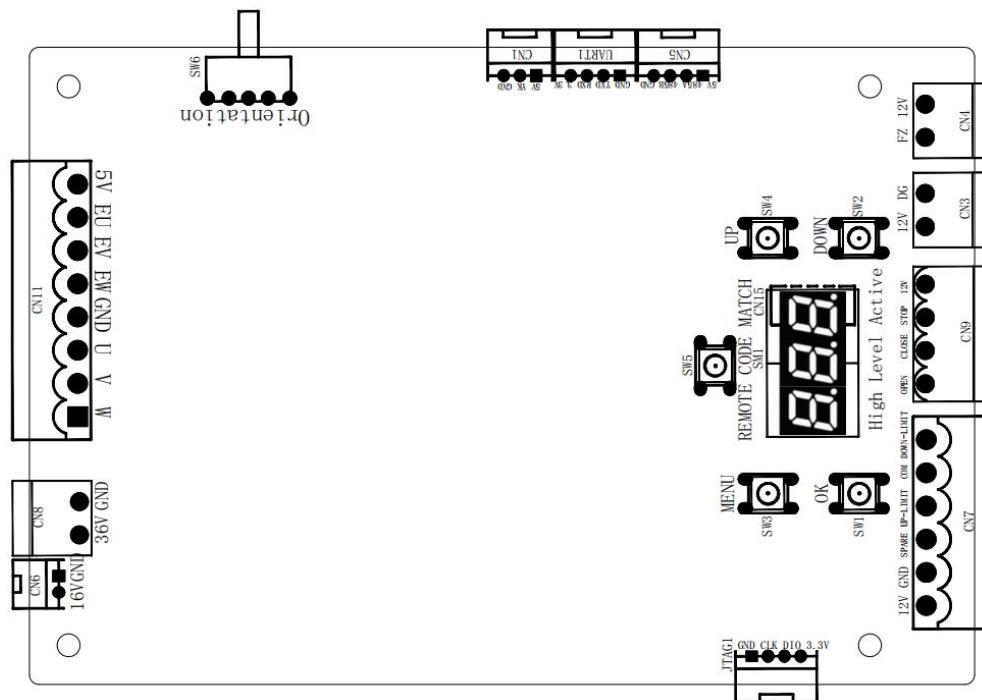
5.1 Wiring and Terminal Definition

5.1.1 Controller Wiring Diagram



Controller Wiring Diagram

5.1.2 Low-Voltage Mainboard Terminal Definition



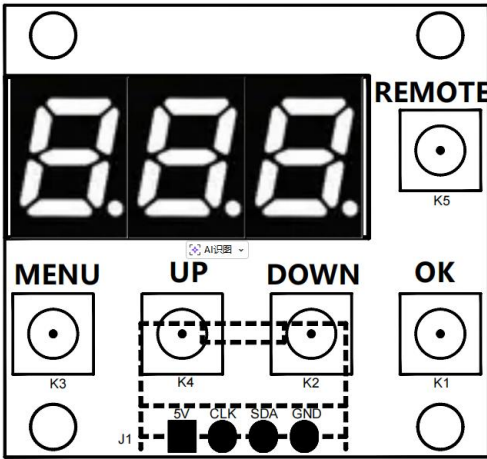
Low-Voltage Mainboard Terminal Definition

Interface Description

SN	Terminal definition	Remark
CN1	5V YK GND	Remote control
CN3	12V DG	Detector
CN4	12V FZ	Anti-hit
CN5	5V 485A 485B GND	RS485communi cation
CN6	16V GND	Charge
CN7	12V GND spare UP-limit COM Down-limit	Limit-switch
CN8	36V GND	Power Supply
CN9	Open Close Stop 12V	Control input
CN11	5V EU EV EW GND U V W	Motor

CN15	GND SDA CLK 5V	Digital-display
UART1	GND TX RX 3.3V	RS232 Communication
JTAG	GND CLK DIO 3.3V	Download
SW6	Forward Reverse	Orientation-dire ction

5.1.3 Display Board Terminal Definition Diagram

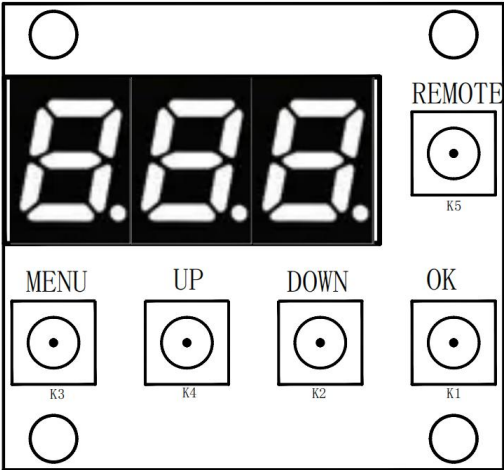


Display Board Terminal Definition Diagram

Interface Description

Ternimal SN	Terminal definition	Remark
J1	5V CLK SDA GND	To Mainboard

5.2 Operation Instructions for the Control Driver



"UP": In normal working mode, this key opens the gate. After entering the setting mode, it is used to navigate up the menu items and increase the set value. In the parameter setting interface, a short press increases the value by 1 each time. Holding it down continuously increases the value to the maximum, then restarts from the minimum.

"DOWN": In normal working mode, this key closes the gate. After entering the setting mode, it is used to navigate down the menu items and decrease the set value. In the parameter setting interface, a short press decreases the value by 1 each time. Holding it down continuously decreases the value to the minimum, then restarts from the maximum.

"MENU": Enters the setting interface and returns to the upper level menu.

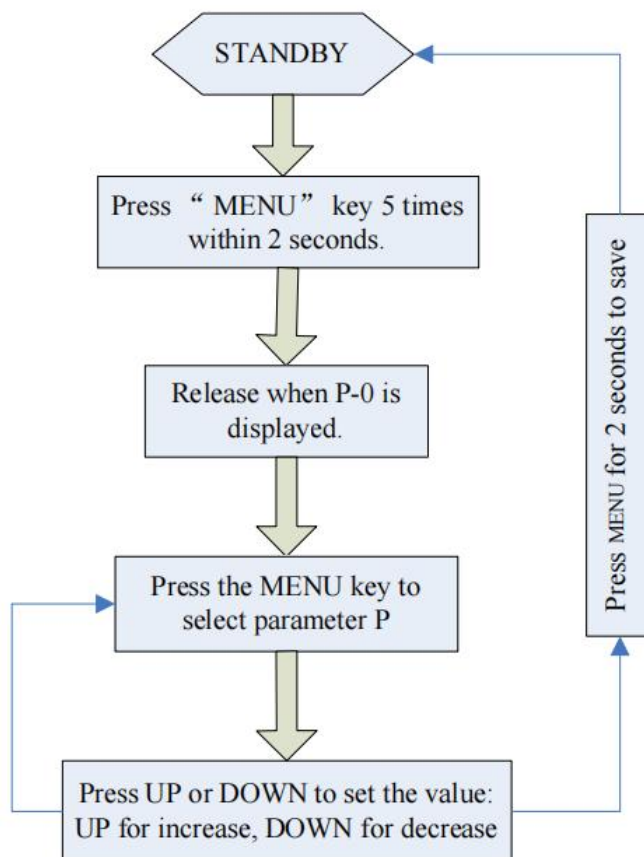
"OK": Confirms the setting value or returns to one level menu.

6 Parameter Setting and Debugging

The opening and closing operation time of the barrier gate and other parameters are set via the setting keys under seven-segment display. The definitions of the keys are shown in the figure below. The settings include parameter setting, one-key restoration and remote control code matching.

6.1 Parameter Setting

Parameter setting method is as follows:



The parameter setting table is as follows:

SN	Parameter	Content	Default	Remark
P Advanced Menu				
1.	P-0	Opening Response Speed: Setting range: 1–200. The larger the value, the faster the response speed (Position Loop Kp)	130	
2.	P-1	Closing Response Speed: Setting range: 1–200. The larger the value, the faster the response speed (Position Loop Kp)	150	
3.	P-2	Opening Position Adjustment: Adjustment mode: jog adjustment. Up $\uparrow$, Arm Vertical Position Adjustment. By short-pressing or long-pressing the Up and Down keys, the arm can be adjusted to the vertical position. Exit and save to confirm the vertical position.	无	
4.	P-3	Closing Position Adjustment: Adjustment mode: jog adjustment. Down $\downarrow$, Arm Horizontal Position Adjustment. By short-pressing or long-pressing the Up and Down keys, the arm can be automatically adjusted to the horizontal position. Exit and save to confirm the horizontal position.	无	
5.	P-4	Opening Increment KP Starting Angle (Opening Deceleration Angle): Setting range: 0–90.0 means no deceleration; 30 means the value starts to decrease or increase when the barrier arm is at the last 30 degrees of travel.	60	
6.	P-5	Closing Decrement KP Starting Angle (Closing Deceleration Angle): Setting range: 0–90.0 means no deceleration; 30 means the value starts to decrease or increase when the barrier arm is at the last 30 degrees of travel.	30	
7.	P-6	Opening Deceleration Time: Setting range: 1–50, representing 0.1–5.0. The larger the value, the smaller the acceleration zone and the larger the deceleration angle; the smaller the value, the larger the acceleration zone and the smaller the deceleration angle.	10, represent 1.0	
8.	P-7	Closing Deceleration Time: Setting range: 1–50, representing 0.1–5.0. The larger the value, the smaller the acceleration zone and the larger the deceleration angle; the smaller the value, the larger the acceleration zone and the smaller the deceleration angle.	25, represent 2.5	
9.	P-8	Opening Slope Adjustment Value: Setting range: 1–500. This controls the acceleration smoothness in the opening deceleration zone. The smaller the value, the less obvious the deceleration effect when entering the deceleration zone (Speed Loop Deceleration Zone Ki).	110	
10.	P-9	Closing Slope Adjustment Value: Setting range: 1–500. Controls the acceleration smoothness in the closing deceleration zone. The smaller the value, the less obvious the deceleration effect when entering the deceleration zone. (Speed Loop Deceleration Zone Ki)	10	
11.	P10	Opening end Force: Setting range: 1–50. Increase this value if the movement is slow or jittery when reaching the end; decrease this value if shaking occurs when reaching the end.	5	
12.	P11	Closing end Force: Setting range: 1–50. Increase this value if the movement is slow or jittery when reaching	8	

		the end; decrease this value if shaking occurs when reaching the end.		
13.	P12	Obstacle Bounce Speed Percentage:Setting range: 0–9. The smaller the value, the more sensitive the bounce;	1	
14.	P13	Obstacle Bounce Response Time:Setting range: 0.0–5.0, where 0.1 represents 100ms.The larger the value, the slower the system response and the less likely false triggering will occur.	0.2	
15.	P14	Motor Maximum Force:Setting range: 10–100.Maximum output force during barrier gate operation (PWM duty cycle).	90	
16.	P15	Auto Close Time interval Without vehicle pass:Setting range: 0–300.If no vehicle is detected by the loop detector for X seconds after opening, the barrier gate will close automatically.For example, 10 means waiting 10 seconds to close automatically; 0 means disabling this function.	0	
17.	P16	Delayed Closing Time after vehicle pass:Setting range: 0–200, unit: second.After the loop detector detects a vehicle passing, the gate will close after a delay of X seconds.The closing speed follows the "Auto Slow Closing Speed	0	
18.	P17	Opening Count:Setting range: 0–3,0: Disabled, 1: Enabled. The gate will close only after the loop detector detects N passages when N times of opening are recorded,2: Anti-tailgating mode,3: counting when the loop detector is triggered	0	
19.	P18	Emergency Stop Buffer Time:Setting range: 0.1–1.0 seconds.The smaller the value, the shorter the braking travel.	0.2	
20.	P19	Emergency Stop Shaft Locking Force:Setting range: 1–40.This is the maximum shaft locking force after emergency stop. The larger the value, the greater the shaft locking force.	10	
21.	P20	Motor Rated Speed:Setting range: 1–60.The motor rated speed equals the set value × 100 rpm	50, represent 5000rpm	
22.	P21	Aging Test:Setting range: 0.0–20.0,0: Aging test disabled.1.0–20.0: After arm run its end, wait X seconds before automatically runing.	0	
23.	P22	Remote Address Coding Mode (20-bit or 16-bit):0: 433 coding (2,4,8);1: 430 coding (30,03,0C);2: 430 coding (0C,30,C0);3: 433 coding (4,1,2);4: 433 coding (8,4,2)	0	
24.	P23	Orientation Current Limit:Setting range: 0–100, which is the maximum current output limit for change giving (PWM duty cycle).	40	
25.	P24	Opening Acceleration Time:Setting range: 1–50, representing 0.1–5.0 seconds.1 unit equals 100 ms. The larger the value, the softer the startup	2, represent 0.2 S	
26.	P25	Closing Acceleration Time:Setting range: 1–50, representing 0.1–5.0 seconds.1 unit equals 100 ms. The larger the value, the softer the startup	2, represent 0.2 S	
27.	P26	Opening end Speed:Setting range: 0–100, unit: 10 rpm.The larger the value, the faster the speed when reaching the end.	5	
28.	P27	Closing end Speed:Setting range: 0–100, unit: 10 rpm.The larger the value, the faster the speed when reaching the end.	5	
29.	P28	Photosensitive Sensitivity:Setting range: 1–33.The larger the value, the earlier the light turns on	25	

30.	P29	Photosensitive Delay Time:Setting range: 1–300. When the photosensitive sensor detects that the light meets the switching condition, the light will be turned on or off after a delay of X seconds	20	
31.	P30	Orientation Speed:Setting range: 1–100. This value represents the orientation speed; the larger the value, the faster the speed (percentage of rated speed).	20	
32.	P31	Motor Orientation Mode:Setting range: 0–2,0: Orientation upward,1: Orientation downward,2: Orientation both upward and downward	2	
33.	P32	Opening Rebound Dead Zone:Setting range: 1–100, unit: degree.Indicates how far from the boundary the gate is during opening.	90	
34.	P33	Closing Rebound Dead Zone:Setting range: 1–100, unit: degree.Distance from the boundary during gate closing.	10	
35.	P34	Barrier Opening End Relay Mode:Setting range: 0–9	6	
36.	P35	Barrier Closing End Relay Mode:Setting range: 0–9	9	
37.	P36	Motor Maximum Protection Duration:Setting range: 1–100, unit: second.Maximum duration for gate opening or closing	20	
38.	P37	Protocol Selection:Setting range: 0–900,0: RS485 protocol,1–900: MQTT protocol upload interval	0	
39.	P38	RS485 Communication ID:Setting range: 1–255	1	
40.	P39	RS485 Baud Rate:Setting range: 0–3,0: 115200 bps,1: 9600 bps,2: 19200 bps,3: 38400 bps	0	
41.	P40	Locking Shaft Force at end:Setting range: 0–50, default locking shaft force at position (PWM duty cycle)	0	
42.	P41	Loop Detector Signal Sensitivity:Setting range: 0.1–10.0, minimum valid duration of the loop detector signal	0.2	
43.	P42	Anti-lift Angle:Setting range: 0–50. When the barrier arm is X° away from the open/close limit position, the limit command will continue to be executed	10	
44.	P43	Maximum E-Stop Locking Duration:Setting range: 0–300 seconds.If the shaft remains locked for more than X seconds after emergency stop, the gate will open.If set to 0, the shaft will stay locked permanently	30	
45.	P44	Delay Time After Obstacle Bounce:Setting range: 0–50. After the obstacle bounce signal of the barrier gate disappears, the gate will continue to close after a delay of X seconds. If set to 0, the gate will not lower after opening fully and will only lower when a command is received.	0	
46.	P45	Disable Port Signal Mapping:Setting range: 0–1,0: No mapping,1: Map to gate open signal	0	
47.	P46	Loop Detector Bypass Angle:Setting range: 0–100. The loop detector signal is not detected after the barrier arm drops below X degrees	20	
48.	P47	Loop Detector Allowed Lowering Angle:Setting range: 0–100. During gate opening, if the loop detector is triggered, the gate will descend only after rising above X degrees.	87	
49.	P48	Remote Control Allowed Lowering Angle:Setting range: 0–100. During gate opening, if receiving close command , then the barrier will descend only after rising above X degrees.	87	
50.	P49	Deceleration Mode:Setting range: 0–10: Standard Mode1: Limit Pressing Mode	0	
51.	P50	Opening Deceleration Stroke:Setting range: 1–60, unit:	25	

		degree, first-stage deceleration stroke for gate opening		
52.	P51	Closing Deceleration Stroke:Setting range: 1–60, unit: degree, first-stage deceleration stroke for gate closing	7	
53.	P52	Gate Opening Acceleration Zone Smoothness:Setting range: 1–999, smoothness of gate opening acceleration.A larger value results in faster acceleration (Speed Loop Acceleration Zone Ki)	15	
54.	P53	Gate Closing Acceleration Zone Smoothness:Setting range: 1–999, smoothness of gate closing acceleration.A larger value results in faster acceleration (Speed Loop Acceleration Zone Ki).	15	
55.	P54	Gate Opening Acceleration Response:Setting range: 1–999, gate opening acceleration response (Speed Loop Kp)	500	
56.	P55	Gate Closing Acceleration Response:Setting range: 1–999, gate closing acceleration response (Speed Loop Kp)	400	
57.	P56	Gate Opening Base Speed:Setting range: 1–100, gate opening base speed (percentage of rated speed)	70	
58.	P57	Gate Closing Base Speed:Setting range: 1–100, gate closing base speed (percentage of rated speed)	70	
59.	P58	Buzzer Switch:Setting range: 0–1,0: Off; 1: On	1	
60.	P59	Anti-freezing Angle:Setting range: 0–90, opening angle after the anti-freezing timer expires	0	
61.	P60	Anti-freezing Time Interval:Setting range: 0–100, unit: minutes.Open and close the gate once when the cumulative duration of no door-opening operation reaches this set time.	0	
62.	P61	Loop Detection Valid Angle:Setting range: 0–90. The loop detector becomes valid when the arm lowers to this angle.90 means valid always.This angle must be greater than the Loop Non-Detection Angle to form a loop detection zone.	90	
63.	P62	Long-time Gate Open Alarm Output Duration:Setting range: 0–60. When the gate open signal lasts for X seconds, the long-time gate open alarm signal will be sent to the background server via the serial port.	0	
64.	P63	Ad Light Control Mode:Setting range: 0–3,0: Light sensor control,1: Time control,2: Light sensor + time control,3: Light sensor on, delayed off	0	
65.	P64	Real-Time Clock Hour Calibration:Setting range: 0–23, Real-Time Clock (RTC) hour calibration (hardware support required)	12	
66.	P65	Real-Time Clock Minute Calibration:Setting range: 0–59, Real-Time Clock (RTC) minute calibration (hardware support required)	0	
67.	P66	Ad Light Turn-On Time:Setting range: 0–235, turn on at 18:30 (hardware support required)	185	
68.	P67	Ad Light Turn-Off Time:Setting range: 0–235, turn off at 6:00 (hardware support required)	60	
69.	P68	Power-off Gate Control Switch:Setting range: 0–20: Disabled1: Open gate on power failure2: Close gate on power failure	0	
70.	P69	Software Current Limit Maximum:Setting range: 0.0–50.0, maximum output current of the power supply (Unit: 1A)	20.0	
71.	P70	Lock Shaft Current at End:Setting range: 0.0–5.0, maximum lock shaft current at in-position (Unit: 1A), used in conjunction with P40.	1.0	
72.	P71	Low Voltage Protection Time:Setting range: 0–100, low voltage protection time (Unit: 0.1 second)	30, represent 3.0	

73.	P72	Low Voltage Protection Trigger Voltage:Setting range: 0–30, low voltage protection trigger voltage (Unit: 1V).The system enters low voltage protection status when the voltage is below this value	14	
74.	P73	Low Voltage Protection Recovery Voltage:Setting range: 0–30, low voltage protection recovery voltage (Unit: 1V).The system enters low voltage recovery state when the voltage is above this value, and resumes normal operation after the protection time elapses.	16	
75.	P74	Orientation Direction:Setting range: 0–1,0: Orientation with motor forward rotation;1: Orientation with motor reverse rotation	0	
76.	P75	Motor Type:Setting range: 0–4	4	
77.	P76	Current Loop KP:Setting range: 1–999	70	
78.	P77	Current Loop KI:Setting range: 1–999	200	
79.	P78	Default C Value: Setting Range: 1-7, 1->3.0M, 2->3.5M, 3->4.0M, 4->4.5M, 5->5.0M, 6->5.5M, 7->6.0M	1	
80.	P79	Seven-Segment Display Mode:The display modes are as follows:0 - Display running status,1 - Display pulse,2 - Display clock,3 - Display count,4 - Display detection,5 - Display software version number	0	
81.	P80	Breathing Light Color Setting:The setting range : 0–6, as detailed below:0- Red,1- Green,2- Blue,3-Red + Green,4-Red + Blue,5- Green + Blue,6- Red + Green + Blue	5	
82.	P81	System minimum current output duration setting.Setting range: 1–999, precision 0.1 second, representing 0.1–99.9 seconds.Protection is triggered when the set current exceeds the value specified in P-82 for this duration	50,represent 5.0	
83.	P82	Minimum Current Protection Threshold:Setting range: 1-999, precision 0.1 ampere, representing 0.1-99.9A	10, represent1.0	
84.	P83	Anti-hit Action:Setting range: 0-1,0-Stop when signal is detected;1-Rise after signal is detected	1	
85.	P84	Anti-Evasion Protection Angle:Associated with P-83.0 – No angle protection;1 – Based on P83, if the tilt angle is less than 5 degrees, the gate bar stops operating and remains stationary. It resumes closing after the vehicle has left	0	
86.	P85	Automatic Slow Closing Speed:When automatically closing the gate without a loop detector, the gate closes at this speed.Setting range: 1–200.A larger value results in faster speed response, while a smaller value results in slower speed response (Position Loop Kp).	30	

6.2 One-key Restore Factory Default Parameters

6.2.1 Hardware Factory Reset

In standby mode, long press the "OK" key until the seven-segment displays "----". Then press the "UP" or "DOWN" key four times, and press the "OK" key again to restore factory settings.

6.3 Remote Control Code Match & Clear

6.3.1 Remote Control Code Match

Press and hold the remote control matching key for 2–3 seconds to enter setting interface. Press any key on the remote control. A short beep indicates successful matching; a long beep indicates the remote control already exists. A maximum of 250 remote controls can be matched.

6.3.2 Remote Control Code Clear

In this interface, press and hold the “UP” key or “DOWN” key to clear all remote control codes.

6.3.3 Exit Remote Control Code Match & Clear

Press the “remote control matching” key to exit remote control learning. The remote control will take effect only after exiting the setting interface.

6.4 Adjustment of Opening Direction

If the opening and closing direction is reversed during debugging, please enter menu D-2 and/or D-3 to set the left/right opening position before proceeding with debugging.

6.5 Aging

6.5.1 Enter Aging Mode

After debugging is completed, the device can be set to aging mode for aging test. Follow the function parameter setting procedure to enter the "P21" setting interface and set the value greater than 1. The gate will enter aging mode, and the interval during aging will be executed according to the set value. The device must remain on the P21 interface during aging.

6.5.2 Exit Aging Mode

Follow the function parameter setting procedure to enter the "P-21" setting interface. Change the value of P-21 to 0 or press the MENU key to exit the current interface; both operations will exit aging mode.

7 Troubleshooting

SN	FAQ	Solution
1	Seven-segment is not working	Check the circuit for any short circuit or open circuit.
2	The barrier always open	1.Check if the open signal always exist. 2.Check if the anti-crush signal is always present.
3	E01	Main motor overcurrent: caused by the motor current exceeding the bearing capacity of the hardware current. Solutions are as follows: 1.Check the balance of the barrier gate's springs. If unbalanced, adjust the spring tension or replace the springs. 2.Inspect the control board.
4	E02	Hall fault 1.Check whether the Hall wiring is normal. 2.Check whether the Hall sensor is faulty.
5	E03	Barrier is stalled or jammed for long-time, this is generally caused by incorrect "up and down limit" settings of the barrier gate. The mechanical limit is positioned ahead of the set opening/closing position, causing the motor to stall before the gate reaches its proper position. Solutions are as follows: 1.Adjust the "opening/closing position" toward the normal operating range. 2.Check if the gate arm is jammed; if so, resolve the jam.
6	E04	Peak Mode,Check whether Peak Mode is activated. This is triggered when the open signal is held for more than 6 seconds. Press the "DOWN" key to clear Peak Mode.
7	E05	Vehicle Detector Protection,The vehicle detector continuously outputs a signal. Check if the vehicle detector is frozen or invalid. Solutions are as follows: If the vehicle detector freezes, press the reset button or power off the detector and restart it. If the vehicle detector is invalid, check for poor contact or damage to the loop coil. If the fault still cannot be resolved, replace the control board to verify normal function.
8	E06	Anti-hit Protection,The anti-hit signal remains continuously active. Check if the anti-hit sensor is always outputting a signal. Solutions are as follows: 1.Check if the anti-hit infrared / radar sensor is blocked by objects. If so, remove the obstacles. If not, check the sensor power supply. If power is normal, the infrared sensor is faulty and should be replaced. 2.Check if the anti-hit pressure wave sensor is under constant pressure. If not, inspect the pressure wave sensor and replace it if defective. 3.If both above are normal, check the control board.
9	E07	Low Voltage,Caused by insufficient power supply voltage to the controller. Solutions are as follows: 1.Check whether the power supply voltage of the controller is normal (under no-load or normal load conditions). If abnormal, replace the power supply. 2.If the power supply is normal, inspect the control board.
10	E08	Memory fault: the mainboard needs to be replaced.

11	E09	Fixed information is invalid.
12	E10	Encoder fault: 1.Check whether the encoder wiring is normal; 2.Check whether the encoder is faulty.
13	E11	Power failure:Check for power supply fault or poor power connection.
15	E12	Rod disengagement: Check if the gate rod has been knocked loose; Check if the gate rod sensor is abnormal.
16	E13	Illegal cabinet door opening: Check if the cabinet door is open or the door sensor is faulty.
17	E14	Illegal rod lifting: Check if someone has forced the gate rod open.
18	E15	Gate closing blocked:Check as follows: 1.Obstacles, pedestrians, vehicles, etc. under the gate arm. 2.Mechanical jamming or obstruction.
19	E16	Synchronization error: Check whether the synchronization cable is securely connected or reversed.
20	E17	Large current protection:Same as the troubleshooting method for E01.

8 After-Sales Service

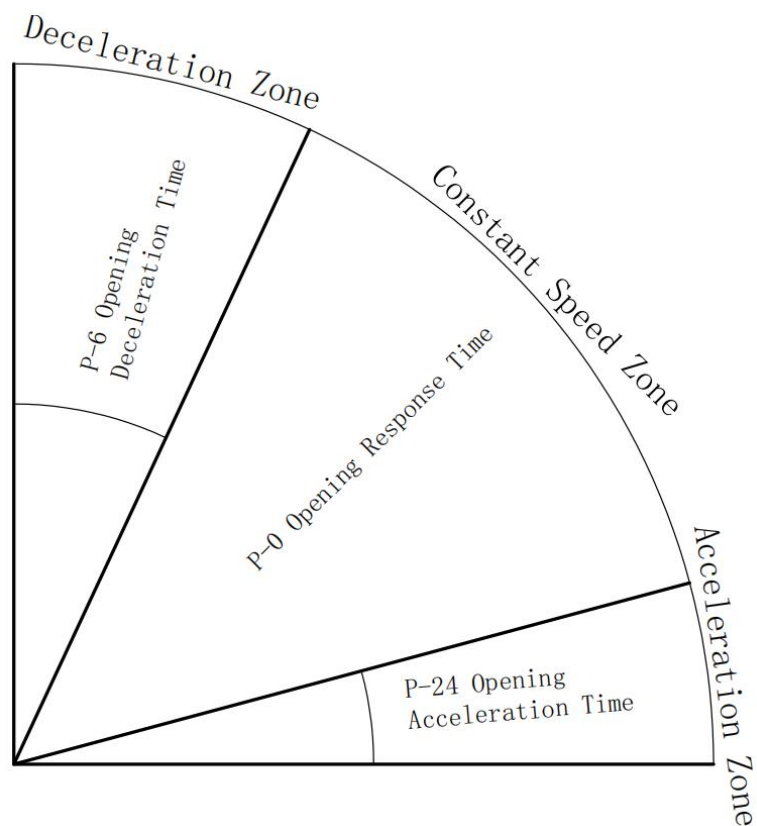
- 1.Regularly check the terminal connections to ensure they are secure.
- 2.Check whether the earthing is in good condition.
- 3.Carefully verify that the power supply is correct before use.

Appendix 1: Correspondence Table of Open/Close Limit Relay Modes

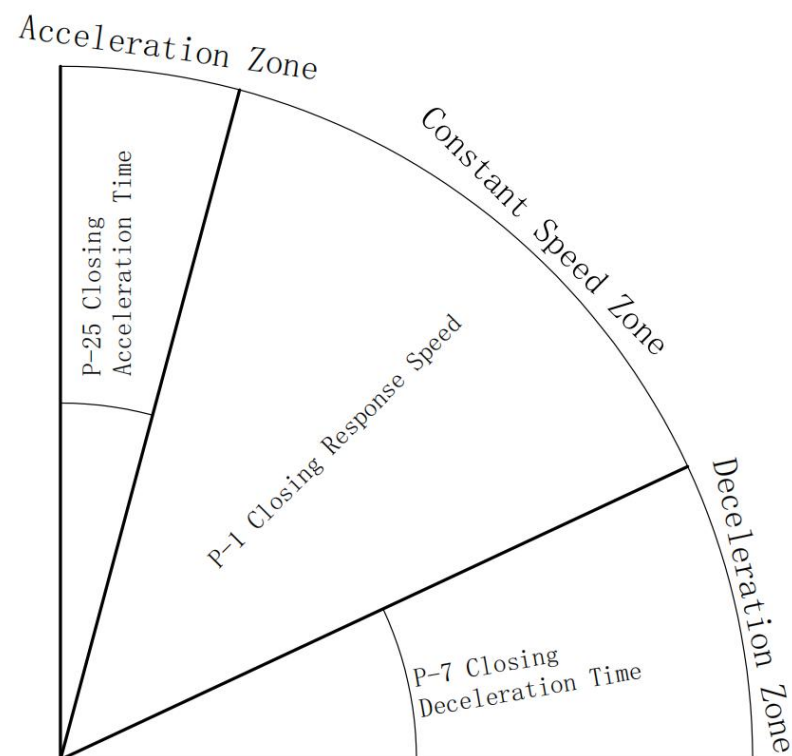
SN	Mode	Spare NC	UP-Limit(P34) NO	COM	DOWN-Limit(P35) NO
1	Advertising Light Mode: 0	Night:Relay close Daytime Relay open			Night:Relay close Daytime Relay open
2	Alarm Mode: 1		After illegal open,Relay close for 500ms,then Relay open		After illegal open,Relay close for 500ms,then Relay open
3	Power supply mode: 2		After arm rising, Relay close,After arm is close, Relay open		After arm rising, Relay close,After arm is close, Relay open
4	Traffic Light 1: 3	After arm is horizontal, Relay close	After arm is open, Relay close,After arm is close, Relay open		After arm is close, Relay close,After arm is open, Relay open
5	Traffic Light 2: 4	After arm is closing,Relay close	After arm rising, Relay close,After arm descending, Relay open		After arm descending, Relay close,After arm rising, Relay open
6	linkage Mode: 5		After receiving open signal,Relay close for 500ms,then Relay open		After receiving close signal, Relay close for 500ms,then Relay open
7	Tri-state Mode: 6		After arm is open,Relay close,other relay open		After arm is close,Relay close,other relay open
8	Pulse Mode: 7		After arm is open,Relay close for 500ms,then Relay open		After arm is close,Relay close for 500ms,then Relay open
9	Motor state: 8		During arm running,Relay close ,other Relay open		During arm running,Relay close ,other Relay open
10	Traffic Light 3: 9		After arm is close,Relay open,other relay close		After arm is open,Relay open,other relay close

Appendix 2: Functional Schematic Diagram

1. Open Schematic Diagram



2. Close Schematic Diagram



3. Position Setting Diagram

